

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – [LIBRARY AI AGENT]

Domain of Project – [Education .]

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Recommended font style and size:

Font Style: Arial

Font Size (Heading): 28

Font Size (Content): 20

Problem Statement -

Problem Statement: Library AI Agent

The Challenge

Students often spend a lot of time searching for the right books and learning resources for their subjects. Traditional library searches may require students to know the exact **book title, author, or keywords**, making it difficult to discover useful resources.

With a large number of books and academic materials available, students may also struggle to identify which resources are most relevant to their **course, syllabus, and current study topic**.

Information about book availability may not always be easy to access, and students may need to manually check whether a book is available or join a waiting list.

The Objective

Build an **AI-powered Library Agent** that understands student queries and provides personalized learning-resource recommendations. The solution should:

- ❖ **Natural Language Search** – Understand student questions and search for relevant books using everyday language.
- ❖ **Personalized Recommendations** – Suggest books based on the student's subject, course, topic, and academic requirements.
- ❖ **Syllabus-Based Matching** – Match study topics with relevant books and learning materials.
- ❖ **Book Availability** – Check whether recommended books are available in the library.
- ❖ **Reservation & Waitlist** – Help students reserve unavailable books or join a waitlist.
- ❖ **Smart Prioritization** – Highlight highly relevant and high-demand resources.
- ❖ **AI Assistance** – Use IBM Granite to understand queries, reason about relevant resources, and generate helpful recommendations.

Goal:

Reduce the time students spend searching and make library resources easier to discover, access, and use.

Proposed Solution -

Proposed Solution: AI-Powered Library Agent

The proposed solution is an AI-powered Library Agent built using **IBM Bob**, **IBM watsonx.ai**, and **IBM Granite** to help students quickly find relevant books and learning resources based on their academic needs.

At its core, the system uses **Natural Language Processing (NLP)** to understand student queries and match them with suitable books and resources from the library database.

The system consists of four key functions:

- ❖ **Book Recommendation Agent** – Recommends relevant books based on the student's subject, topic, course, and learning requirements.
- ❖ **Library Search Agent** – Understands natural-language queries and searches the library database for suitable resources.
- ❖ **Availability Agent** – Checks real-time availability of books and identifies whether a book is available, issued, or in high demand.
- ❖ **Reservation Agent** – Assists students with reserving unavailable books and joining waitlists.

The solution can also provide **personalized recommendations** by considering the student's course, study topic, and academic requirements.

IBM Bob is used to orchestrate the agent workflow, while **IBM Granite** provides natural-language understanding, reasoning, and personalized responses. **IBM Cloud/watsonx.ai** supports deployment and scalable AI services.

Overall, the Library AI Agent makes the library search process **faster, smarter, and more personalized**, helping students find the right learning materials efficiently.

Technology Used

- **IBM Bob platform** -For designing and orchestrating multi-agent workflows visually
- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **RAG**– RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database** (FAISS/Chroma) – For RAG-based nutritional data retrieval

Required Files created using IBM bob -

The screenshot displays the IBM Bob IDE interface. The main editor window shows the file `library_tools.py` with the following content:

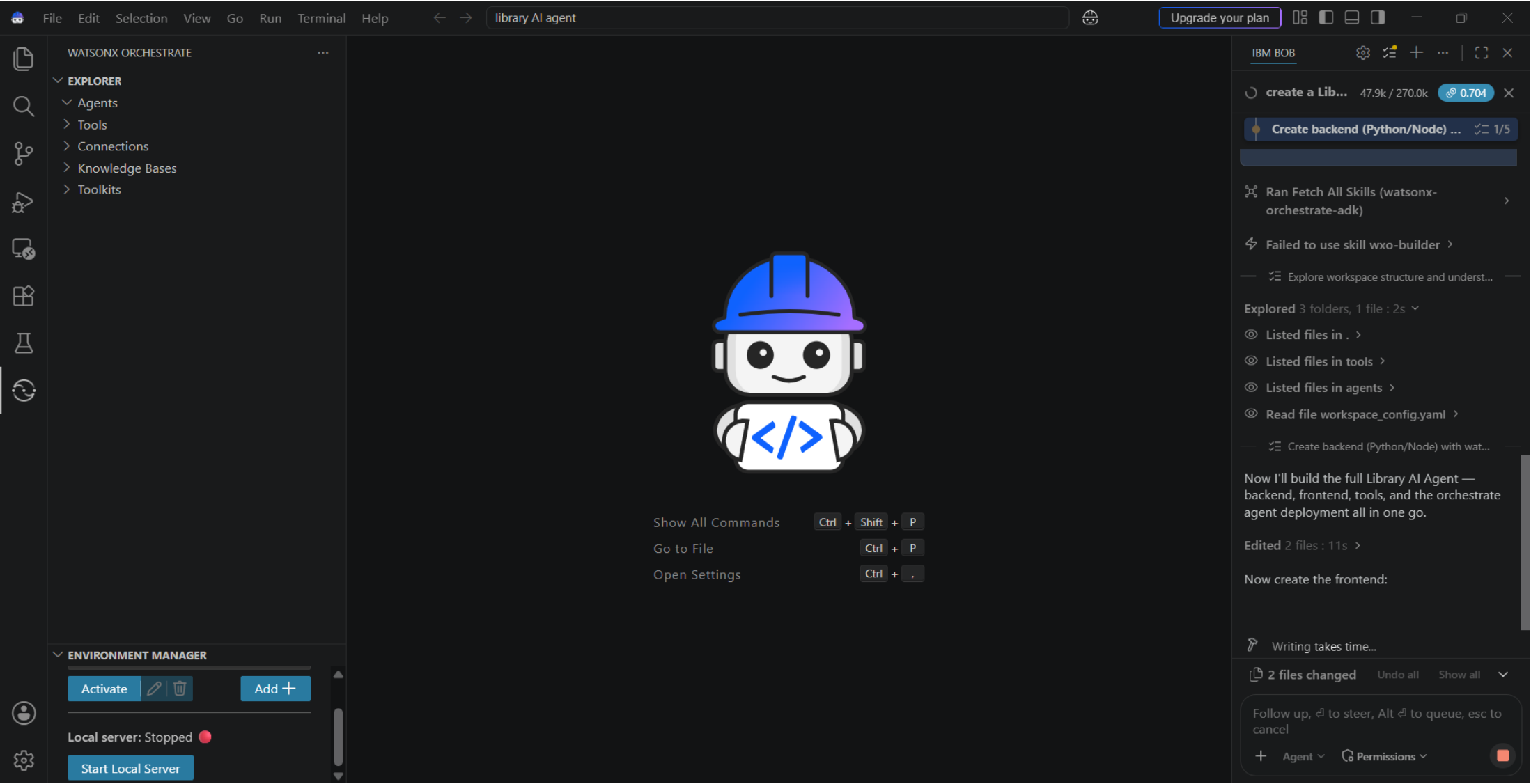
```
1  """
2  Library AI Agent - watsonx Orchestrate Tools
3  Four tools exposed to the orchestrate platform:
4  1. search_library_books - full-text search across the catalog
5  2. check_book_availability - check copies in stock for a specific title
6  3. reserve_library_book - place a reservation or waitlist entry
7  4. get_ai_book_recommendations - call watsonx LLM for personalised suggestions
8  """
9
10 import json
11 import requests
12 from datetime import datetime, timedelta
13 from ibm_watsonx_orchestrate.agent_builder.tools import tool, ToolPermission
14
15 # --- Config ---
16 BACKEND_URL = "http://localhost:5000/api" # adjust if deployed elsewhere
17 WATSONX_URL = "https://au-syd.ml.cloud.ibm.com/ml/v1/text/generation?version=2023-05-29"
18 WATSONX_API_KEY = "2mmLKomgg--0HR5yh5Y9IWMDf-WqNA_SkZxGYub-cZ5B"
19 MODEL_ID = "meta-llama/llama-3-3-70b-instruct"
20 PROJECT_ID = "addfe882-3b61-4b3b-ab5d-d1bf25cbb5dd"
21 IAM_TOKEN_URL = "https://iam.cloud.ibm.com/identity/token"
22
23 _token_cache: dict = {"token": None, "expires_at": None}
24
25
26 def _get_iam_token() -> str:
27     now = datetime.utcnow()
28     if _token_cache["token"] and _token_cache["expires_at"] and _token_cache["expires_at"] > now:
29         return _token_cache["token"]
30     resp = requests.post(
31         IAM_TOKEN_URL,
32         headers={"Content-Type": "application/x-www-form-urlencoded"},
33         data={"grant_type": "urn:ibm:params:oauth:grant-type:apikey", "apikey": WATSONX_API_KEY},
34         timeout=30,
35     )
36     resp.raise_for_status()
```

The left sidebar shows the **WATSONX ORCHESTRATE** project structure with folders for **Agents**, **Tools**, **Connections**, **Knowledge Bases**, and **Toolkits**. Below this is the **ENVIRONMENT MANAGER** section, which includes buttons for **Activate**, **Add +**, and **Start Local Server**. The status indicates **Local server: Stopped**.

The right sidebar shows the **IBM BOB** interface. It includes a **create a Libr...** button with a progress indicator (60.9k / 270.0k) and a cost of 1.29. Below this, it lists tasks: **Create and register tools in wat...** (3/5). The main text area on the right provides instructions: **Now I'll build the full Library AI Agent — backend, frontend, tools, and the orchestrate agent deployment all in one go.** It also shows **Edited 2 files : 11s** and **Now create the frontend:** with a link to **Wrote file frontend/index.html**. Below that, it says **Now create the watsonx Orchestrate tools:** with a list of tasks: **Create and register tools in watsonx Orc...**, **Create frontend (HTML/CSS/JS) for the L...**, and **Create and register tools in watsonx Orc...**. At the bottom, it says **Now write the WxO tool file:**.

The bottom status bar shows the following information: **ADK: 2.16.1**, **Ln 1, Col 1**, **Spaces: 4**, **UTF-8**, **Python 3.13.9 (venv: venv)**, and **Bob Settings**.

IBM Bob Screenshots:



WATSONX ORCHESTRATE

EXPLORER

Agents

Tools

Connections

Knowledge Bases

Toolkits

ENVIRONMENT MANAGER

Activate

Add +

Local server: Stopped

Start Local Server

library AI agent

Upgrade your plan

IBM BOB

create a Library ... 68.3k / 270.0k 6.11

All tasks completed! 5/5

The MCP server is pointing to a different environment key. Let me check the existing tools and agents, then write the agent spec file and create the agent directly:

Called MCP 2 tools : 2m 11s

The orchestrate environment needs to be activated first. I'll write the full agent spec file and a complete setup script so everything can be deployed with a single command:

Wrote file agents/library-ai-agent.yaml

Now write the deploy script and a requirements file for the tools:

Edited 2 files : 5s

Now write a comprehensive README:

Wrote file README.md

Now let's validate the backend by checking for syntax errors:

Ran 3 commands

8 files changed Undo all Show all

Follow up or start new task Ctrl+N

+ Agent Permissions

README.md

library-ai-agent.yaml

README.md

Architecture

36

Quick Start

51

3. Deploy to watsonx Orchestrate

57

Then deploy tools + agent

58

python deploy.py

59

60

61

62

63

Configuration

64

Variable | Value |

65

---|---

66

WATSONX_URL | https://au-syd.ml.cloud.ibm.com/ml/v1/text/generation?version=2023-05-29 |

67

MODEL_ID | meta-llama/llama-3-3-70b-instruct |

68

PROJECT_ID | addfe882-3b61-4b3b-ab5d-d1bf25cbb5dd |

69

API_KEY | Set in backend/app.py and tools/library_tools.py |

70

71

72

73

74

API Endpoints

75

Method | Endpoint | Description |

76

---|---|---

77

GET | /api/health | Health check |

78

GET | /api/books | List all books (filter: subject, available) |

79

GET | /api/books/search?q=<query> | Full-text search |

80

GET | /api/books/<id> | Get single book |

81

POST | /api/books/<id>/reserve | Reserve or join waitlist |

82

POST | /api/recommendations | AI-powered recommendations |

83

POST | /api/chat | Conversational AI chat |

84

GET | /api/subjects | List all subjects |

85

GET | /api/stats | Library statistics |

86

87

88

89

90

watsonx Orchestrate Tools

ADK: 2.16.1

Ln 10, Col 4

Spaces: 4

UTF-8

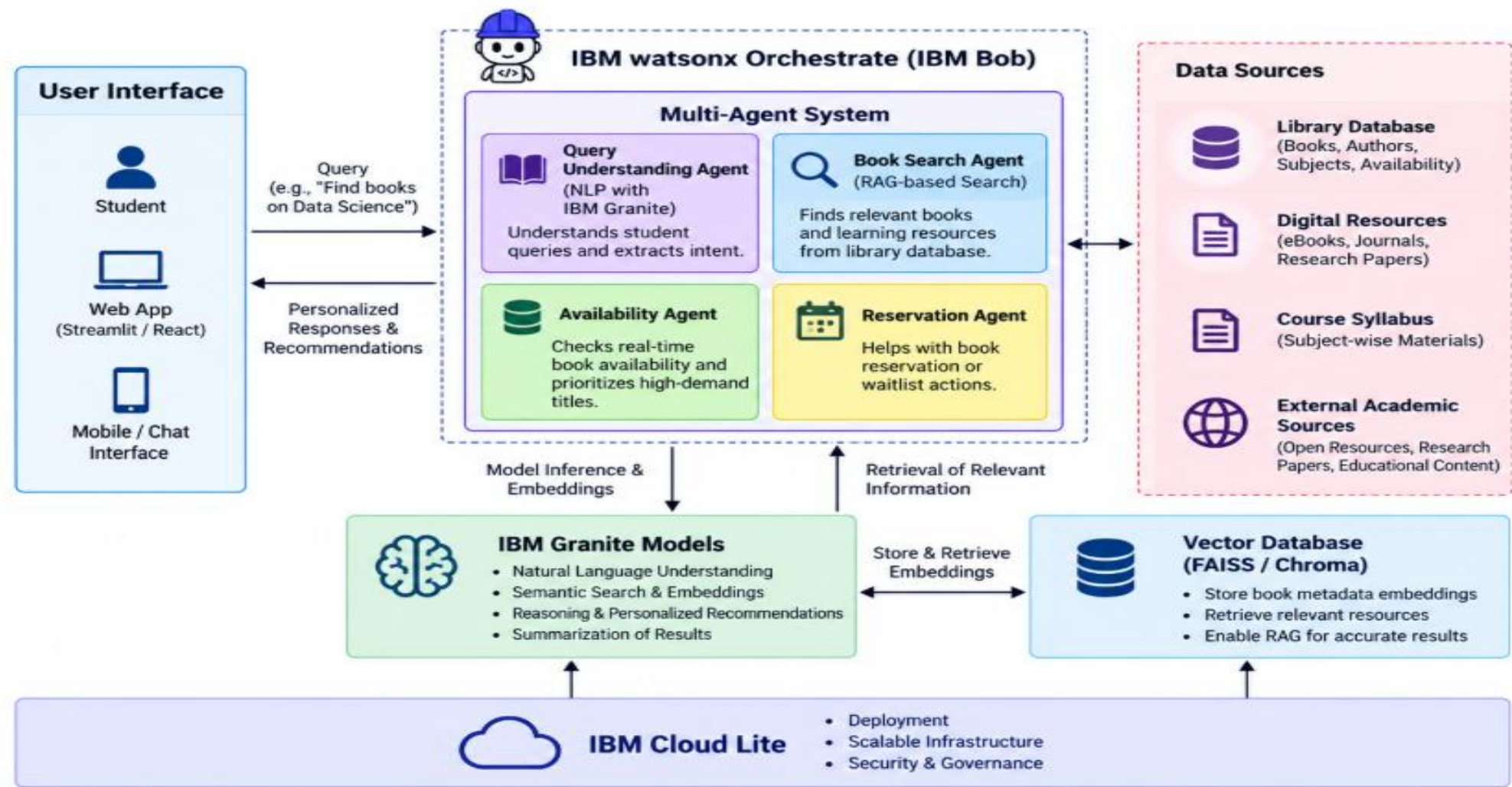
LF

Markdown

Bob Settings

Architecture blueprint for IBM bob:

Architectural Blueprint – Library AI Agent



Role of Agentic AI in the solution

Role of Agentic AI in Library AI Agent

Agentic AI enables the Library AI Agent to work as an **intelligent and autonomous library assistant** rather than a simple search system. Using **IBM Bob**, **IBM watsonx.ai**, and **IBM Granite**, the system coordinates multiple specialized agents to understand student requirements and provide relevant learning resources.

Each agent performs a specific task while working together:

Query Understanding Agent – Understands the student's natural-language question and identifies the subject, topic, or resource required.

Library Search Agent – Searches the library database and finds relevant books and learning materials.

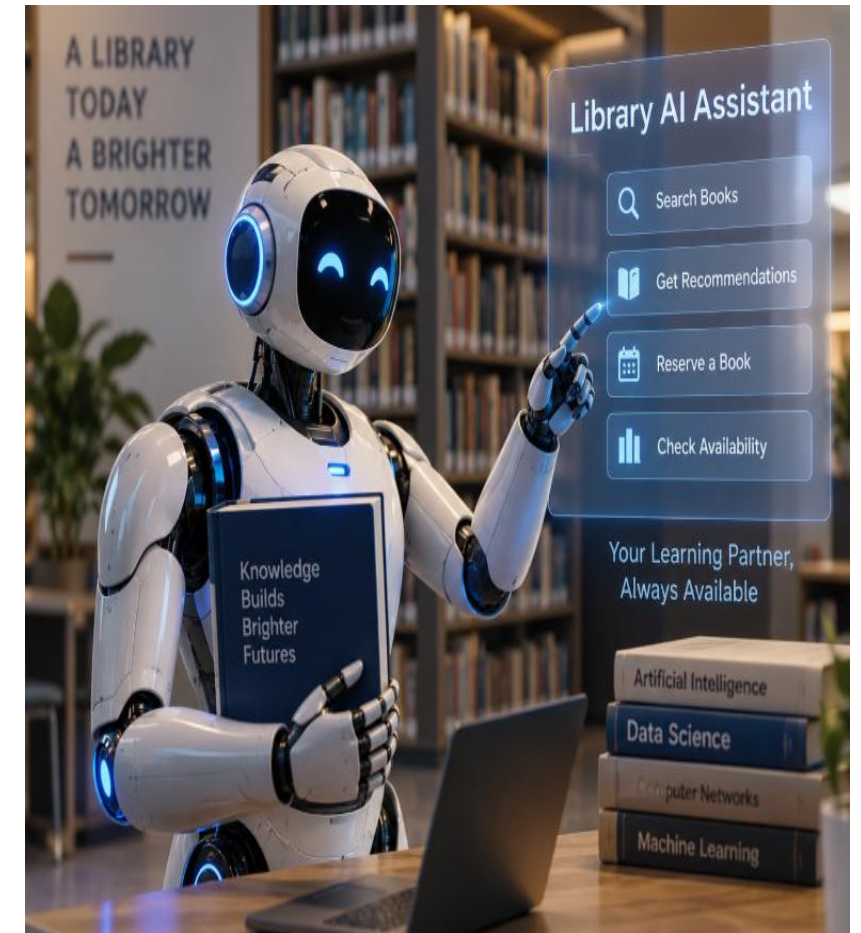
❖ **Availability Agent** – Checks the current availability and status of recommended books.

❖ **Reservation Agent** – Helps students reserve available books or join a waitlist for unavailable resources.

Agentic AI provides **context awareness** by considering the student's course, syllabus, subject, and study requirements when generating recommendations.

It also enables **autonomous decision-making**, allowing the agents to determine which action is required—searching for a book, checking availability, or assisting with a reservation.

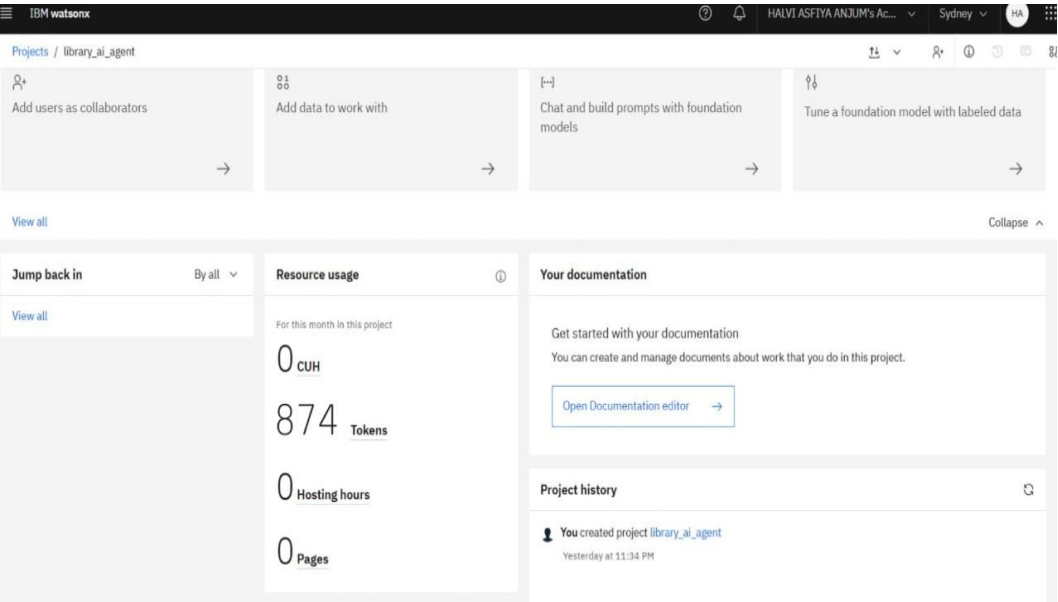
Overall, Agentic AI makes the Library Agent proactive, personalized, and efficient, helping students find and access the right academic resources with less manual searching.



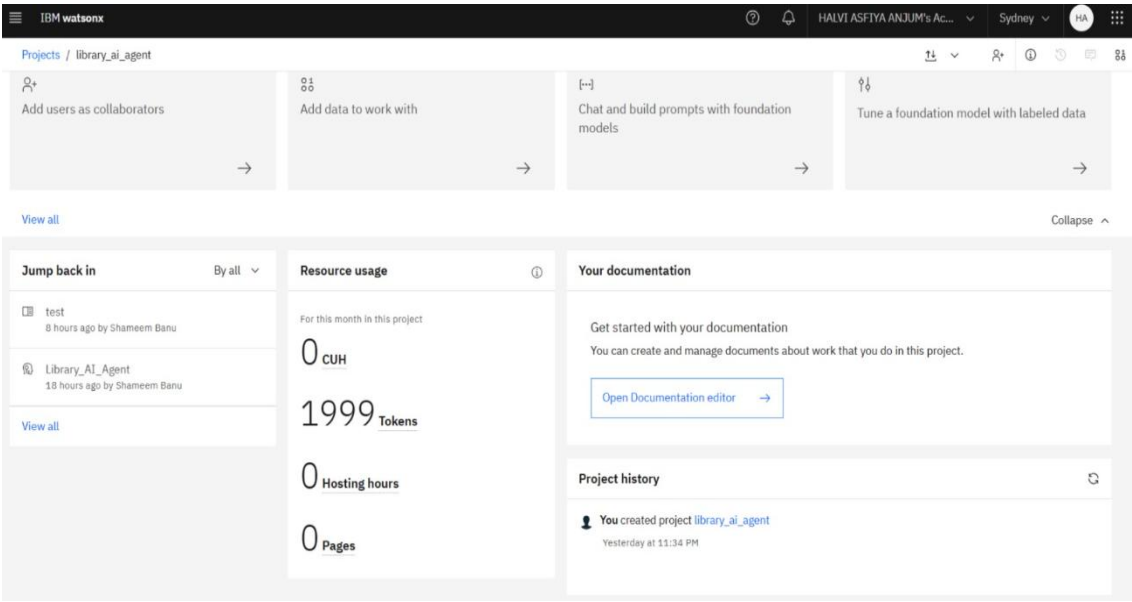
Token Usage

11,761 token are used

Before Prompt



After the Prompt



Project Output Screenshot

 Library AI Agent

[Catalog](#) [Recommend](#) [Chat](#) [Reserve](#)

TOTAL TITLES
15

AVAILABLE
12

SUBJECTS
9

TOTAL COPIES
59

 All Subjects

Available Only

15 Books

Introduction to Algorithms
by Cormen et al. · 2022
[Computer Science](#)
● 3 of 5 available
★★★★★ 4.8
[Reserve](#)

Clean Code
by Robert C. Martin · 2008
[Software Engineering](#)
● 2 of 3 available
★★★★★ 4.7
[Reserve](#)

The Pragmatic Programmer
by Hunt & Thomas · 2019
[Software Engineering](#)
● All copies checked out
★★★★★ 4.9
[Reserve](#) [Waitlist](#)

Artificial Intelligence: A Modern Approach
by Russell & Norvig · 2020
[Artificial Intelligence](#)
● 1 of 4 available
★★★★★ 4.6
[Reserve](#)

Deep Learning
by Goodfellow et al. · 2016
[Machine Learning](#)

Design Patterns
by Gang of Four · 1994
[Software Engineering](#)

Computer Networks
by Tanenbaum & Wetherall · 2010
[Networking](#)

Operating System Concepts
by Silberschatz et al. · 2018
[Operating Systems](#)

WHAT ARE YOU STUDYING?

machine learning algorithms

SUBJECT AREA

Machine Learning

ACADEMIC LEVEL

Undergraduate

[🔥 Get Recommendations](#)

****Book Recommendations:****

1. **"Deep Learning"** by Goodfellow et al. - This book is a comprehensive introduction to deep learning techniques, a crucial part of machine learning. It's suitable because it directly addresses the student's interest in machine learning algorithms.
2. **"Machine Learning with Python"** by Andreas Müller - This book is ideal for practical implementation of machine learning concepts using Python. It's suitable because it combines theoretical knowledge with practical application, which is essential for understanding machine learning algorithms.
3. **"Introduction to Algorithms"** by Cormen et al. - While broader than just machine learning, this book provides a solid foundation in algorithms, which is fundamental to machine learning. It's suitable because understanding general algorithms is crucial for grasping machine learning algorithms.

****Why Each Book is Suitable:****

- **Deep Learning** and **Machine Learning with Python** are directly related to the subject area, providing specific insights into machine learning algorithms.
- **Introduction to Algorithms** offers a foundational understanding necessary for advanced machine learning concepts.

****Suggested Reading Order:****

****Suggested Reading Order:****

1. Start with **"Introduction to Algorithms"** to establish a strong foundation in algorithms.
2. Then, read **"Machine Learning with Python"** to apply algorithmic knowledge to machine learning problems.
3. Finally, dive into **"Deep Learning"** for specialized knowledge on deep learning techniques.

****Additional Study Tips:****

- Practice coding with libraries like TensorFlow or PyTorch to apply theoretical knowledge.
- Engage with online resources and communities, such as Kaggle, to stay updated on the latest developments in machine learning.
- Work on projects that integrate machine learning algorithms to reinforce learning.

Remember, machine learning is a practical field, so applying theoretical knowledge is key. Good luck with your studies!

Matching Books

Introduction to Algorithms

by Cormen et al.

● 3 copies available

Reserve

Deep Learning

by Goodfellow et al.

● 2 copies available

Reserve

Machine Learning with Python

● 2 copies available

Reserve

Project Output Screenshot

 **Library AI Agent**

 Catalog  Recommend  Chat  Reserve



Hello! I'm your Library AI Assistant. I can help you find books, check availability, make reservations, and suggest resources based on your studies. What can I help you with today?

I need help with machine learning. What books do you recommend?



For machine learning, I recommend the following books from our library: 1. "Machine Learning" by Andrew Ng - This book provides a comprehensive introduction to machine learning, covering topics such as supervised and unsupervised learning, neural networks, and deep learning. 2. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville - This book is a detailed guide to deep learning, covering topics such as convolutional neural networks, recurrent neural networks, and generative models. 3. "Pattern Recognition and Machine Learning" by Christopher Bishop - This book provides a broad introduction to machine learning, covering topics such as probability theory, linear regression, and support vector machines. All three books are currently available in our library. Would you like to check them out or reserve them for later? Additionally, I can recommend some study materials and online resources to supplement your learning. We also have a quiet study room on the second floor where you can work on your machine learning projects. Would you like more information on these resources?

Ask me anything about the library...

Send

QUICK QUESTIONS

 Programming books available?

 ML book recommendations

 AI & Data Science books

 How to reserve a book?

 Library subjects

STATUS

 15 titles

 12 available

 9 subjects

 59 total copies

Project Output Screenshot

 Library AI Agent

 Catalog  Recommend  Chat  Reserve

 Reserve a Book

SEARCH FOR BOOK

Type to search...

SELECTED BOOK

Database System Concepts

YOUR NAME

livia spencer

STUDENT ID

e.g. S12345

Confirm Reservation

Novelty and Uniqueness

The proposed **Library AI Agent** combines **Agentic AI, IBM Granite, NLP, RAG, and personalization** to create a smart and efficient academic resource assistant.

- ❖ **Multi-Agent Intelligence** – Multiple specialized agents collaborate to understand queries, search books, check availability, and manage reservations.
- ❖ **Natural Language Interaction** – Students can ask questions in simple language instead of using exact book titles or complicated search keywords.
- ❖ **Personalized Recommendations** – Recommends books and resources according to the student's **subject, course, syllabus, and study topic**.
- ❖ **RAG-Based Search** – Retrieves relevant information from the library database and academic resources before generating responses, helping provide more relevant results.
- ❖ **Real-Time Availability** – Checks the current status of books and identifies available, issued, reserved, or high-demand titles.
- ❖ **Smart Reservation & Waitlist** – Assists students in reserving books and joining waitlists when resources are unavailable.
- ❖ **Context-Aware Assistance** – Understands the student's academic requirements and uses that context to provide more useful recommendations.
- ❖ **Time-Saving & Efficient** – Reduces manual searching and helps students quickly discover the right learning materials.

Overall, the **Library AI Agent** is more than a traditional library search system—it acts as a **personalized, proactive, and intelligent learning-resource assistant**.

Git Hub Link

https://github.com/halvi-786/library_ai_agent

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Future Scope

1. Voice & Multilingual Support

The Library AI Agent can support **voice-based queries and multiple languages**, making it easier for students to search and interact with the system.

2. Smart Library Integration

Integration with **RFID and barcode systems** can provide real-time information about book issue, return, location, and availability.

3. Personalized Recommendations

The agent can analyze **student interests, course requirements, and previous searches** to provide more accurate and personalized book recommendations.

4. Digital Academic Resources

The system can be expanded to recommend **eBooks, journals, research papers, online courses, and other digital learning materials**.

5. Predictive Resource Management

AI can analyze borrowing patterns to identify **high-demand books** and help libraries plan inventory and resource allocation more efficiently.

Future Vision: Develop the Library AI Agent into a **complete intelligent library platform** that provides personalized learning support and improves access to academic resources.

Certificates Earned:

Please add your credly certificate(Getting Started with Artificial Intelligence)



Certificates Earned:

Add your IBM BOB Certificate



Thank You!

Thank you for your time and interest.